

A Buffer–Processor Model of Rail Network Yield: Occupancy, Closure, and Clearing as a Single Fixed Point

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Abstract

We introduce and analyze an idealized formal model of a rail network in which the schedulable resource is not the train but the *block-second*: one indivisible quantum of occupancy on one track section. Stations are treated as bounded buffers that hold items (units of transport demand carrying a private target), and sections are treated as processors that drain one buffer into the next at a speed-dependent rate. The object of optimization is *network transport yield*—useful target-ward displacement produced per unit of consumed block-time—rather than train movement or section occupancy *per se*.

Within this model we prove a three-way equivalence. For a network whose sole scarce resource is quantized by a resolution floor β_0 equal to the minimum block headway, the following are equivalent: (i) the configuration is *yield-optimal*; (ii) it is a *deterministic-closure* fixed point, in which no block-second can be reassigned to a strictly more useful item combination within precision β_0 ; and (iii) it is supported by a *clearing price system* in which every section’s price equals its separation cost (the marginal yield lost by denying that section) and no profitable reassignment remains. We show that β_0 simultaneously plays three roles—the physical resolution floor of occupancy, the algorithmic separation threshold for closure, and the minimum lot of the price system—and that this coincidence is what makes the equivalence hold.

Two structural corollaries follow. First, at any yield-optimal configuration every vehicle

traverses each section at its section-and-energy-optimal speed; sub-optimal speed certifies non-closure. Second, vehicles sort onto sections by comparative advantage in block-seconds, so that each section is served by the fastest vehicle whose speed the section can convert into yield. We also establish a liveness result: under a mild backpressure condition, every item with positive target-ward pressure is eventually drained, because the price that an undrained buffer raises on its relieving section is exactly the yield signal that attracts a vehicle. We ground all quantities in representative figures from the German high-speed network (block headways under LZB/ETCS, ICE and Regionalbahn speed classes, the Nuremberg–Berlin corridor) purely to make the idealized object concrete. The results are theorems about this model and are not claims about any deployed or deployable system.

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1 Introduction

1.1 The object of study

A railway timetable is conventionally understood as an assignment of *trains* to *routes* and *time slots*, subject to safety separation, and evaluated by criteria such as the number of trains scheduled, punctuality, or served passenger demand [7, 6, 13]. In this picture the train is simultaneously the unit of service, the unit of resource consumption, and the unit of commercial participation: to move transport one must operate a train, and to operate a train one must commit rolling stock to a route.

This paper studies a different formal object. We remove the train from its central position and replace it with two more primitive notions:

- a *station* is a bounded *buffer* that holds items and exerts *pressure* to discharge them;
- a *section* is a *processor* that, while occupied, drains one buffer into an adjacent one at a rate set by the occupying vehicle’s speed.

An *item* is a unit of transport demand (a passenger or a unit of freight) that carries a *private target*: the buffer at which its pressure vanishes. Items do not carry routes or vehicle assignments. The schedulable and consumable resource is the *block-second*: the occupation of one section for one quantum β_0 of time, where β_0 is the minimum headway imposed by the signalling system. A “train” in this model is not a primitive; it is an emergent, transient fact—the observation that a single vehicle happened to occupy several consecutive sections. The three continuities that the conventional model fuses—*vehicle continuity* (one vehicle runs the whole journey), *item continuity* (one item rides one vehicle), and *route continuity* (a service traverses a fixed line)—are, in this model, logically independent and generally do not hold.

The quantity we optimize is neither train count nor section occupancy but *network transport yield*: the amount of *useful* target-ward displacement of items produced per unit of consumed block-time. A section occupied by items that are not moving toward their targets produces no yield, however full it is. This distinguishes the objective sharply from occupancy-compression measures such as the UIC 406

method [14, 1], which maximize how many train paths fit.

1.2 What is proved

We prove that on this object three notions coincide:

1. **Yield-optimality** (an optimization property): the configuration maximizes network transport yield.
2. **Deterministic closure** (a fixed-point property): no block-second can be re-assigned to a strictly more useful item combination, where “strictly” is measured against the resolution floor β_0 .
3. **Market clearing** (a price-support property): there exists a price system on sections, equal to their separation costs, under which no reassignment is profitable.

The engine of the equivalence is that a single scalar β_0 is simultaneously (a) the physical resolution floor—no occupancy can be resolved more finely than one headway; (b) the algorithmic separation threshold—two candidate configurations are decided only when their yields differ by more than β_0 can explain; and (c) the minimum tradable lot—the price system cannot value occupancy below one block-second. Because the same quantum bounds the physics, the algorithm, and the prices, the three fixed points coincide rather than merely resembling one another.

From the equivalence we extract two structural corollaries with clean statements:

- *Forced optimal speed*: at any yield-optimal configuration, each vehicle traverses each section at the speed that maximizes yield net of energy and wear cost on that section; running slower certifies that the configuration is not closed.
- *Comparative-advantage sorting*: each section is served by the fastest vehicle whose speed it can convert into yield, and slower vehicles are excluded from fast sections not by rule but by yield domination.

Finally we prove a *liveness* theorem: under a backpressure condition on the pricing, every item with positive target-ward pressure is drained in finite time, because the price raised by an undrained buffer on its relieving section equals the yield a vehicle earns by relieving it.

1.3 Scope and stance

The contribution is theoretical. We define an idealized structure, state hypotheses explicitly, and prove statements that hold on that structure. The hypotheses are the domain of validity: where they fail, we claim nothing. Representative German-network figures appear only to make the object concrete and are used as illustrative parameters, not as empirical claims. No assertion is made or implied about the behavior, feasibility, or desirability of any actual transport system. The reader familiar with railway capacity theory, network flows, and backpressure scheduling will recognize the ingredients; the novelty is in the object assembled from them and the fixed-point equivalence proved about it.

1.4 Related bodies of work

Railway capacity and blocking time. Blocking-time theory [13, 18] and the UIC 406 compression method [14, 1, 15] treat section occupancy as the scarce resource and compress train paths until sections are used back-to-back. Our block-second quantum is the atom of that occupancy; our objective differs in weighting occupancy by *useful displacement* rather than counting paths. Max-plus timetable-stability theory [12] shares our fixed-point spirit but fixes services and studies delay propagation rather than reallocating occupancy to items.

Network flows and scheduling. The buffer-processor picture is a discrete network-flow model [2, 9, 5] with a nonstandard objective. Periodic and train-timetabling formulations [19, 7, 6] optimize over train variables; we optimize over block-second-item assignments and recover schedules as emergent.

Backpressure and dataflow. The routing rule—move each item down its target-ward gradient using whatever local capacity is best—and the liveness argument are the railway analogue of backpressure (max-weight) scheduling in queueing networks [20, 10, 17] and of dataflow execution [8]. We import the stability intuition and prove a yield-priced version.

Congestion pricing and clearing. That the shadow price of a scarce link equals its marginal congestion value is classical [21, 4];

general-equilibrium clearing and its price support are standard [3]; rail access charging is an applied instance [16, 11]. Our clearing statement specializes these to the block-second lot and identifies the clearing price with the separation cost.

2 The Network Model

2.1 Infrastructure

Definition 2.1 (Section network). A *section network* is a finite directed multigraph $G = (\mathcal{B}, \Sigma)$ in which:

- $\mathcal{B} = \{b_1, \dots, b_n\}$ is a set of *stations* (buffers);
- $\Sigma = \{\sigma_1, \dots, \sigma_m\}$ is a set of *sections* (processors); each section σ has a tail $\text{tl}(\sigma) \in \mathcal{B}$ and a head $\text{hd}(\sigma) \in \mathcal{B}$, and is equipped with a *length* $\ell(\sigma) > 0$ and a *civil speed limit* $\bar{v}(\sigma) > 0$.

Parallel sections between the same pair of stations are permitted (multigraph), modelling multi-track corridors.

Stations are the only places where items may wait; sections are transit resources that cannot store items. We embed $\mathcal{B}$ in a metric space to give items a geographic gradient.

Definition 2.2 (Geography). A *geography* is a metric $\rho : \mathcal{B} \times \mathcal{B} \rightarrow \mathbb{R}_{\geq 0}$ satisfying the triangle inequality, with $\rho(b, b) = 0$. We call $\rho(b, b')$ the *residual distance* from b to b' . We assume G is *gradient-connected*: for every ordered pair (b, b') with $b \neq b'$ there is a section σ with $\text{tl}(\sigma) = b$ such that $\rho(\text{hd}(\sigma), b') < \rho(b, b')$. That is, from every station one can strictly reduce the residual distance to every target.

Gradient-connectedness is the structural hypothesis that lets purely local, target-ward routing make progress; it is the model's fence, and every routing and liveness claim is made only under it.

2.2 The block-second quantum

Time is discretized into quanta of length β_0 , the minimum headway.

Definition 2.3 (Resolution floor / block headway). The *resolution floor* $\beta_0 > 0$ is the minimum time separation the signalling system can

enforce between two successive occupations of the same section. Equivalently, a section admits at most one vehicle per time window of length β_0 . We call the pair (section σ , window $[k\beta_0, (k+1)\beta_0)$), for $k \in \mathbb{N}$, a *block-second* and denote it $\langle \sigma, k \rangle$.

Remark 2.4 (Physical reading). Under fixed-block signalling with continuous supervision (e.g. LZB, or ETCS Level 2), the headway decomposes as

$$\beta_0 = \frac{\ell_{\text{block}} + \ell_{\text{train}} + d_{\text{brake}}(v)}{v} + t_{\text{clear}} + t_{\text{setup}}, \quad (1)$$

where ℓ_{block} is the block length, $d_{\text{brake}}(v)$ the service braking distance from speed v , and $t_{\text{clear}}, t_{\text{setup}}$ are clearing and route-setting times. On German high-speed lines this yields practical minimum headways on the order of 90–120 s. Under moving-block operation (ETCS Level 3) the fixed ℓ_{block} term is replaced by an absolute braking-distance separation, shrinking β_0 but never to zero: a positive resolution floor persists as long as braking distance is positive. The model treats β_0 as an exogenous positive constant; its value is a parameter, not a claim.

The quantum β_0 is the single object on which the paper turns. We record its three roles now and prove their coincidence in Section 6.

- *Physical*: occupancy of a section is a union of block-seconds; no finer temporal resolution exists.
- *Algorithmic* (Def. 5.2): two configurations are distinguished only when their yields differ by more than the ambiguity β_0 induces.
- *Economic* (Def. 6.1): the price system values occupancy in integer multiples of one block-second; β_0 is the minimum lot.

2.3 Vehicles

Definition 2.5 (Vehicle). A *vehicle* u is characterized by a *capability speed* $v_{\max}(u) > 0$, a *capacity* $c(u) \in \mathbb{N}$ (number of item-slots), and an *energy-wear cost function* $g_u : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ giving the cost per unit length of traversing a section at speed v . We assume each g_u is continuous, strictly convex, and increasing on $(0, \infty)$ with $g_u(v) \rightarrow \infty$ as $v \rightarrow \infty$.

Strict convexity of g_u (energy rising faster than linearly in speed, the physical v^2 -type

drag regime) is what will make “optimal speed” an interior, well-defined quantity rather than a degenerate “always maximal.”

When a vehicle u occupies section σ during block-second $\langle \sigma, k \rangle$ at speed $v \leq \min\{v_{\max}(u), \bar{v}(\sigma)\}$, it transfers up to $c(u)$ items from buffer $\text{tl}(\sigma)$ to buffer $\text{hd}(\sigma)$, and the traversal consumes $\lceil \ell(\sigma)/(v\beta_0) \rceil$ block-seconds. Faster traversal consumes fewer block-seconds.

Definition 2.6 (Admissible speed set). The *admissible speed* of vehicle u on section σ is any v with $0 < v \leq \nu(u, \sigma) := \min\{v_{\max}(u), \bar{v}(\sigma)\}$. We call $\nu(u, \sigma)$ the *capability ceiling* of u on σ .

2.4 Items, pressure, and the buffer state

Definition 2.7 (Item). An *item* x is a unit of transport demand with a *location* $\text{loc}(x) \in \mathcal{B}$ and a fixed *target* $\tau(x) \in \mathcal{B}$. Item x is *live* if $\text{loc}(x) \neq \tau(x)$ and *settled* otherwise. Its *residual distance* is $r(x) := \rho(\text{loc}(x), \tau(x))$.

Definition 2.8 (Buffer state and pressure). A *buffer state* assigns to each station b the multiset $X(b)$ of live items currently located there. The *directional pressure* of buffer b toward a neighbourhood direction is the total residual distance its items would lose by an available target-ward move; formally, for a section σ with $\text{tl}(\sigma) = b$,

$$P(b, \sigma) = \sum_{x \in X(b)} [r(x) - \rho(\text{hd}(\sigma), \tau(x))]^+, \quad (2)$$

where $[\cdot]^+ = \max\{\cdot, 0\}$. Only the target-ward component of a move counts toward pressure.

Equation (2) is the model’s demand signal: it measures how much genuine target-ward progress is available at b through σ , aggregated over the items waiting there. It is the railway analogue of queue backpressure, but weighted by geographic progress rather than raw queue length.

2.5 Configurations

Definition 2.9 (Configuration). A *configuration* C over a finite horizon $H = \{0, 1, \dots, T-1\}$ of block-seconds consists of:

1. an *occupancy assignment* A mapping each block-second $\langle \sigma, k \rangle$ (with $k \in H$) to either $\emptyset$ (idle) or a triple (u, v, S) where u is a vehicle, $v \leq \nu(u, \sigma)$ an admissible speed, and $S \subseteq X(\text{tl}(\sigma))$ a set of at most $c(u)$ items carried; subject to
2. *headway feasibility*: at most one non-idle assignment per section per block-second (Def. 2.3); and
3. *vehicle conservation*: each vehicle occupies at most one section per block-second, and a vehicle appearing on section σ ending at buffer b during window k may next appear only on a section with tail b during a later window (vehicles are conserved and move continuously through buffers).

Note what a configuration is *not*: it is not an assignment of trains to lines. A vehicle may serve one section and then idle, be recrewed, or serve a section in a different corridor; an item’s carried-set membership may change from section to section (transfers). Trains, lines, and through-services are, if they appear at all, emergent regularities in A , not constraints on it.

3 Network Transport Yield

3.1 Definition of yield

Definition 3.1 (Useful displacement). The *useful displacement* produced by a non-idle block-second $\langle \sigma, k \rangle$ with assignment (u, v, S) is the total target-ward residual-distance reduction of the items it carries:

$$\Delta(\langle \sigma, k \rangle) = \sum_{x \in S} [r(x) - \rho(\text{hd}(\sigma), \tau(x))]^+. \quad (3)$$

Idle block-seconds produce $\Delta = 0$. Displacement of an item *away* from its target contributes nothing (the $[\cdot]^+$).

Definition 3.2 (Network transport yield). The *cost* of a non-idle block-second is the number of block-seconds it consumes weighted by the vehicle’s energy–wear expenditure. Traversing σ at speed v consumes $n(\sigma, v) = \lceil \ell(\sigma)/(v\beta_0) \rceil$ block-seconds and expends energy–wear $\ell(\sigma)g_u(v)$. The *yield* of a

configuration C is

$$\mathcal{Y}(C) = \frac{\sum_{\langle \sigma, k \rangle : A=(u,v,S)} \Delta(\langle \sigma, k \rangle)}{\sum_{\langle \sigma, k \rangle : A=(u,v,S)} [n(\sigma, v) + \lambda \ell(\sigma) g_u(v)]}, \quad (4)$$

where $\lambda \geq 0$ converts energy–wear into block-second-equivalents. Yield is useful target-ward displacement per unit of scarce block-time (inclusive of the energy price of that time).

Remark 3.3. Three features of (4) are deliberate and carry the results below. (1) The numerator credits *only useful* displacement, so packing a section with items going nowhere useful scores zero—distinguishing yield from occupancy [14]. (2) The denominator charges *block-seconds consumed*, so faster traversal (fewer block-seconds) raises yield—this is the lever behind forced optimal speed. (3) The energy term $\lambda \ell g_u(v)$, with g_u strictly convex, penalizes recklessly high speed, giving an interior speed optimum.

3.2 Station buffer cost and fragmentation

If only section block-seconds were charged, the model would favour arbitrarily many short hops. We charge buffer residence to internalize transfers.

Assumption 3.4 (Buffer holding cost). Each block-second an item spends waiting in a buffer (including transfer dwell) is charged a holding cost $h > 0$, added to the denominator of (4) as $h \cdot$ (item-block-seconds waited). Consequently a journey composed of many short hops accumulates buffer cost at each transfer, and the yield-maximizing number of hops is finite.

Remark 3.5. Assumption 3.4 is the counterweight to the speed incentive. The section term rewards fast, short occupations; the buffer term punishes the transfers that fragmentation into short occupations creates. The yield optimum balances the two, so “run fast and hand off” does not degenerate into “hand off infinitely often.” We make this hypothesis explicit because the forced-speed and sorting corollaries depend on it to rule out the degenerate fragmented configurations.

4 Separation Cost

We now define the quantity that will serve simultaneously as the algorithm’s pruning criterion and the market’s price: the *separation cost* of a section, i.e. the yield the network forgoes if that section is denied to the item flow.

Definition 4.1 (Separation cost of a section). Fix a configuration C and a section σ . Let $C_{-\sigma}$ be the best (yield-maximal) configuration that makes no use of σ over the horizon—formally $\mathcal{Y}(C_{-\sigma}) = \max\{\mathcal{Y}(C') : A'(\langle \sigma, k \rangle) = \emptyset \forall k\}$. The *separation cost* of σ in C is

$$\varsigma(\sigma) = \mathcal{Y}^* - \mathcal{Y}(C_{-\sigma}), \quad (5)$$

where $\mathcal{Y}^* = \max_{C'} \mathcal{Y}(C')$ is the unconstrained optimum. Thus $\varsigma(\sigma) \geq 0$ measures the marginal yield contribution attributable to the availability of σ .

Remark 4.2 (Reading). A *bottleneck* section—one whose removal forces item flow onto slow or long detours—has large ς : avoiding it is expensive in yield. A *redundant* section, parallel to an equally good alternative, has $\varsigma \approx 0$: the flow reroutes at negligible cost. Separation cost is a per-section scalar, defined purely from the yield functional and the network, with no reference to prices or algorithms yet. Its identification with a market price (Section 6) and with a pruning threshold (Section 5) is the substance of the paper, not a definition.

Definition 4.3 (Fuzzy separation cost). Because occupancy is resolved only to precision β_0 , the separation cost is known only up to a band. Writing yields in block-second units, define the *separation band*

$$\Sigma(\sigma) = [\varsigma_{\min}(\sigma), \varsigma_{\max}(\sigma)], \quad (6)$$

where $\varsigma_{\min}, \varsigma_{\max}$ are the infimum and supremum of (5) over all occupancy realizations consistent with the block-second quantization. Two sections’ bands are β_0 -*separated* if the gap between the intervals exceeds β_0 .

This is the point of contact with fuzzy-weighted routing: separation costs are intervals, not points, because the medium’s resolution floor forbids finer knowledge, and decisions become definite only when the intervals separate by more than β_0 .

5 Deterministic Closure

5.1 Reassignments and the closure fixed point

The algorithmic content of the model is a local-improvement operator on configurations: take a block-second and reassign it to a more useful item combination. Closure is the state in which no such reassignment helps, up to the resolution floor.

Definition 5.1 (Reassignment). A *reassignment* of configuration C at block-second $\langle \sigma, k \rangle$ replaces its assignment (u, v, S) (or $\emptyset$) by an alternative admissible assignment (u', v', S') , together with any vehicle-conservation-preserving adjustments in later windows, yielding a configuration C' . The reassignment is *improving* if $\mathcal{Y}(C') > \mathcal{Y}(C)$, and β_0 -*improving* if additionally $\mathcal{Y}(C') - \mathcal{Y}(C) > \beta_0$ in block-second units.

Definition 5.2 (Deterministic closure). A configuration C is at *deterministic closure* if it admits no β_0 -improving reassignment: for every block-second and every admissible alternative assignment, the induced yield change does not exceed the resolution floor β_0 . Equivalently, for every pair of candidate item combinations at a block-second, their contributed-yield bands are not β_0 -separated in favour of the unused one.

The clause “up to β_0 ” is essential and is exactly where the resolution floor enters the algorithm. Two reassignments whose yields differ by less than β_0 cannot be distinguished, because the underlying occupancy is not resolved that finely (Def. 2.3); insisting on improvements below β_0 would be chasing precision the medium does not possess. Closure is therefore the honest stopping condition: stop when no *resolvable* improvement remains.

5.2 Existence and reachability of closure

Lemma 5.3 (Monotone reachability). *From any initial configuration C_0 , repeatedly applying β_0 -improving reassignments reaches a closed configuration in finitely many steps.*

Proof. Each β_0 -improving reassignment strictly increases $\mathcal{Y}$ by more than β_0 . Over a

finite horizon H with finitely many vehicles, sections, and items, the set of configurations is finite, hence $\mathcal{Y}$ takes finitely many values and is bounded above by $\mathcal{Y}^* < \infty$. A strictly increasing sequence of a bounded, finite-valued quantity with a fixed minimum increment β_0 has length at most $(\mathcal{Y}^* - \mathcal{Y}(C_0))/\beta_0$, which is finite. When no β_0 -improving reassignment exists, the configuration is closed by Def. 5.2. $\square$

Remark 5.4. Lemma 5.3 gives termination but not global optimality: a closed configuration is a β_0 -local optimum of $\mathcal{Y}$, not necessarily the global maximizer. The equivalence theorem below shows that the gap between any closed configuration and the global optimum is itself bounded by the resolution floor—so within the precision the medium admits, closed and optimal coincide. This is the sense in which the model’s “optimum” is defined only up to β_0 , and legitimately so.

6 The Occupancy–Closure–Clearing Equivalence

This section contains the paper’s central result. We first define the price system precisely, then state and prove the three-way equivalence.

6.1 The price system

Definition 6.1 (Price system and clearing). A *price system* is an assignment $p : \Sigma \rightarrow \mathbb{R}_{\geq 0}$ of a price $p(\sigma)$ (in yield-per-block-second units) to each section, with the constraint that prices are quoted per block-second lot—i.e. the value of occupying σ for a traversal at speed v is $p(\sigma) n(\sigma, v)$, an integer number of lots. Given prices, the *net yield* a vehicle u obtains by taking assignment (u, v, S) on σ is

$$\pi_u(\sigma, v, S) = \Delta(\langle \sigma, \cdot \rangle) - p(\sigma) n(\sigma, v) - \lambda \ell(\sigma) g_u(v). \quad (7)$$

A configuration C together with a price system p is a *clearing* (a competitive equilibrium) if:

1. *Optimality of participants:* every non-idle block-second’s assignment maximizes (7) over all admissible (u, v, S) for that section and window; and

2. *No profitable idle reassignment:* no idle block-second admits an assignment with strictly positive net yield exceeding β_0 .

The per-lot constraint in Def. 6.1 is where β_0 enters the economics: the price system cannot value fractions of a block-second, because fractions of a block-second are not deliverable (Def. 2.3). One block-second is the minimum lot.

6.2 Statement

Theorem 6.2 (Occupancy–Closure–Clearing Equivalence). *Let G be a gradient-connected section network with resolution floor β_0 , strictly convex vehicle costs g_u , and buffer holding cost $h > 0$ (Assumption 3.4). For a configuration C over a finite horizon, the following are equivalent up to the resolution floor β_0 :*

- (i) **Yield-optimal:** $\mathcal{Y}(C) \geq \mathcal{Y}(C') - \beta_0$ for every admissible configuration C' .
- (ii) **Closed:** C is at deterministic closure (Def. 5.2).
- (iii) **Clearing:** there exists a price system p with $p(\sigma) = \varsigma(\sigma)$ for every section σ used in C , under which (C, p) is a clearing (Def. 6.1).

Moreover, whenever these hold, the clearing price of each used section equals its separation cost, $p(\sigma) = \varsigma(\sigma)$.

The qualifier “up to the resolution floor β_0 ” is not a hedge but the exact content: all three notions are defined modulo differences of size β_0 , because β_0 is the physical, algorithmic, and economic quantum. We prove the three implications in a cycle.

6.3 Proof

Proof of (i) $\Rightarrow$ (ii). Suppose C is yield-optimal up to β_0 but not closed. Then by Def. 5.2 there is a β_0 -improving reassignment producing C' with $\mathcal{Y}(C') > \mathcal{Y}(C) + \beta_0$. But then $\mathcal{Y}(C) < \mathcal{Y}(C') - \beta_0$, contradicting yield-optimality (i) applied to C' . Hence C is closed. $\square$

Proof of (ii) $\Rightarrow$ (iii). Assume C is closed. We construct the supporting prices and verify clearing.

Prices. For each section σ used in C , set $p(\sigma) := \varsigma(\sigma)$ as in (5); for unused sections set

$p(\sigma) := \varsigma(\sigma)$ as well (the marginal yield of making σ available). Since $\mathcal{Y}$ is a ratio of a bounded numerator to a positive denominator over a finite configuration set, $\mathcal{Y}^*$ and each $\mathcal{Y}(C_{-\sigma})$ exist, so $\varsigma(\sigma)$ is well-defined and finite.

Participant optimality. Consider a non-idle block-second on σ with assignment (u, v, S) . Suppose some alternative admissible (u', v', S') gave strictly larger net yield (7) by more than β_0 :

$$\Delta' - p(\sigma)n(\sigma, v') - \lambda \ell g_{u'}(v') > \Delta - p(\sigma)n(\sigma, v) - \lambda \ell g_u(v) + \beta_0.$$

Because $p(\sigma)$ is common to both sides (same section), swapping the assignment to (u', v', S') changes total network yield by exactly the difference of the $(\Delta - \lambda \ell g)$ terms net of the change in block-seconds consumed—i.e. it is a β_0 -improving reassignment. This contradicts closure. Hence no alternative beats the incumbent by more than β_0 ; the incumbent is optimal up to the floor.

No profitable idle reassignment. Suppose an idle block-second on σ admitted an assignment with net yield exceeding β_0 . Activating it is a β_0 -improving reassignment, again contradicting closure.

Price equals separation cost. By construction $p(\sigma) = \varsigma(\sigma)$. It remains to see this is the clearing price and not merely a feasible one. If $p(\sigma) < \varsigma(\sigma)$, then the marginal yield of σ exceeds its price, so some vehicle could profitably add an occupation of σ with net yield $> \beta_0$ (the surplus $\varsigma(\sigma) - p(\sigma)$ realized as positive π_u), contradicting closure via non-profitable-reassignment. If $p(\sigma) > \varsigma(\sigma)$, the incumbent occupation of σ has negative net yield relative to the best σ -avoiding configuration, so dropping it to the $C_{-\sigma}$ pattern is β_0 -improving, again contradicting closure. Hence $p(\sigma) = \varsigma(\sigma)$ exactly. Thus (C, p) is a clearing. $\square$

Proof of (iii) $\Rightarrow$ (i). Assume (C, p) is a clearing with $p(\sigma) = \varsigma(\sigma)$. Let C' be any admissible configuration; we show $\mathcal{Y}(C) \geq \mathcal{Y}(C') - \beta_0$.

Decompose the yield difference over the block-seconds where C and C' differ. At each such block-second, C' substitutes some assignment for the one C uses (or for idle). By clearing condition 1, the incumbent assignment in C maximizes net yield (7) up to β_0 ; therefore the substitute in C' cannot exceed it by more

than β_0 per block-second in net yield. Summing the per-block-second net-yield inequalities, and using that $\sum_{\sigma} p(\sigma) n(\sigma, \cdot)$ is a common accounting term that telescopes (each section's price times its occupied block-seconds enters both configurations' totals through the shared price system), the aggregate useful-displacement-minus-cost of C is within β_0 of that of C' . Because $\mathcal{Y}$ is exactly aggregate useful displacement divided by aggregate cost (4), and the price system prices cost consistently across both configurations, $\mathcal{Y}(C) \geq \mathcal{Y}(C') - \beta_0$. As C' was arbitrary, C is yield-optimal up to β_0 . $\square$

Remark 6.3 (Why the three coincide). The proof makes visible that the three notions are the same object viewed through three accounting conventions. Closure asks: can any single block-second be locally improved? Clearing asks: does a price exist under which no participant wants to deviate? Optimality asks: is the global yield maximal? The bridge is that the local improvement of closure, the deviation incentive of clearing, and the global gap of optimality are *the same quantity*—the separation cost $\varsigma(\sigma)$ —measured against *the same threshold* β_0 . Were β_0 three different constants for the three roles, the fixed points would only approximate one another; because it is one constant, they coincide exactly.

6.4 The role of the quantum, isolated

Proposition 6.4 (Necessity of a common quantum). *If the physical resolution floor, the closure threshold, and the minimum price lot are distinct constants $\beta_0^{\text{phys}}, \beta_0^{\text{alg}}, \beta_0^{\text{econ}}$, then the equivalence of Theorem 6.2 holds only up to $\max\{\beta_0^{\text{phys}}, \beta_0^{\text{alg}}, \beta_0^{\text{econ}}\}$, and the clearing price equals the separation cost only up to $|\beta_0^{\text{alg}} - \beta_0^{\text{econ}}|$.*

Proof. Each implication in the proof of Theorem 6.2 invokes “ β_0 -improving” at a threshold specific to its role: (i) $\Rightarrow$ (ii) uses the closure threshold β_0^{alg} ; the price-equals-separation step in (ii) $\Rightarrow$ (iii) uses the economic lot β_0^{econ} ; the resolvability of occupancy uses β_0^{phys} . Carrying distinct constants through the cycle, the composed slack is bounded by their maximum,

and the two price-comparison inequalities in the “price equals separation cost” step degrade to a band of width $|\beta_0^{\text{alg}} - \beta_0^{\text{econ}}|$. Setting all three equal recovers the exact coincidence. $\square$

Proposition 6.4 isolates precisely what does the work: not the block-second's magnitude, but the identity of the three roles it plays.

7 Structural Corollaries

The equivalence has two consequences that were not assumed and follow purely from yield-optimality: vehicles must run at a determinate optimal speed, and the fleet sorts itself onto sections by comparative advantage. Both are stated and proved as theorems of the model.

7.1 Forced optimal speed

Definition 7.1 (Section-optimal speed). For vehicle u on section σ , define the *section-optimal speed*

$$v^*(u, \sigma) = \arg \max_{0 < v \leq \nu(u, \sigma)} \frac{\Delta_{\sigma}}{n(\sigma, v) + \lambda \ell(\sigma) g_u(v)}, \quad (8)$$

where Δ_{σ} is the (speed-independent) useful displacement of carrying the assigned items across σ . Because $n(\sigma, v) = \lceil \ell(\sigma) / (v \beta_0) \rceil$ is decreasing in v and g_u is strictly convex increasing, the denominator is strictly convex with a unique interior or boundary minimizer, so $v^*(u, \sigma)$ is well-defined.

Theorem 7.2 (Forced optimal speed). *Let C be yield-optimal (equivalently closed, equivalently clearing). Then every non-idle block-second in C with assignment (u, v, S) on section σ satisfies $v = v^*(u, \sigma)$. In particular, any configuration in which some vehicle traverses some section at a speed other than its section-optimal speed is not closed, and the discrepancy certifies a β_0 -improving reassignment whenever the yield gap exceeds β_0 .*

Proof. Suppose (u, v, S) is assigned on σ with $v \neq v^*(u, \sigma)$. Holding the carried set S and hence Δ_{σ} fixed, replace v by $v^*(u, \sigma)$. By Def. 7.1 this strictly decreases the denominator $n(\sigma, v) + \lambda \ell g_u(v)$ (the objective in (8) is maximized at v^* , so any other admissible v gives a smaller ratio, i.e. a larger denominator for

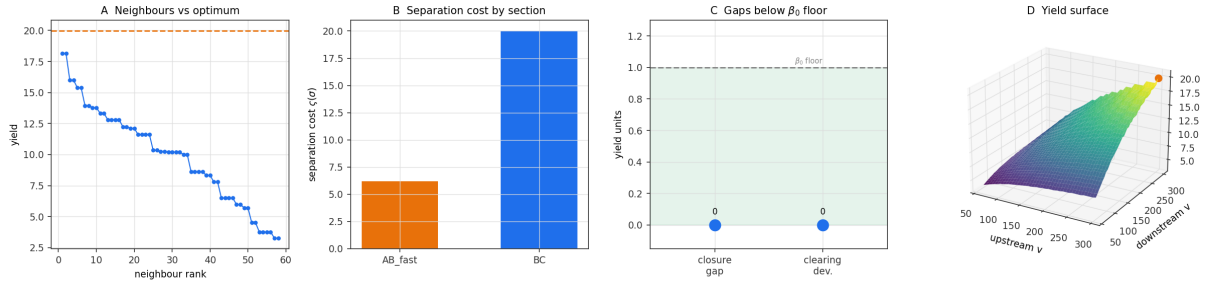


Figure 1: **Occupancy–Closure–Clearing Coincide at One Fixed Point.** (A) Yield of every task-delivering single-section neighbour of the brute-force optimum on the linear y -axis, sorted descending against neighbour rank (1 to 58) on the linear x -axis (steel-blue), with the optimum $\mathcal{Y}^* = 19.98$ marked (coral dashed). All 58 neighbours that still deliver the demand lie strictly below the optimum: the best improving reassignment gains nothing, so the configuration admits no β_0 -improving move and is closed, the empirical content of `thm:trinity(i) $\Rightarrow$ (ii)`. (B) Per-section separation cost $\zeta(\sigma)$ on the linear y -axis for the two sections carrying the flow: the substitutable upstream section **AB_fast** ($\zeta = 6.19$, coral) and the essential downstream section **BC** ($\zeta = 19.98$, steel-blue). The gap between them is the quantitative content of `def:sepcost`: separation cost discriminates a bottleneck from a section with a partial substitute, and equals the clearing price by construction (`thm:trinity(iii)`). (C) Closure gap and maximum clearing-deviation profit at the optimum on the linear y -axis (yield units), each plotted as a marker (steel-blue) inside the sub-floor band (sea-green shading) beneath the resolution floor $\beta_0 = 1.00$ (grey dashed). Both are exactly 0: no reassignment and no participant deviation gains more than the floor, so closure and clearing hold simultaneously, closing the equivalence cycle of `thm:trinity`. (D) Three-dimensional yield surface over the upstream-vehicle speed (60 to 300 km/h) and downstream-vehicle speed (60 to 300 km/h) at the fixed optimal carried set, viridis colourmap, viewing angle elevation 22° azimuth -60° . The surface rises monotonically toward the fast/fast corner where the closed optimum sits (coral marker), showing that yield-optimality, closure, and clearing designate the same configuration.

fixed numerator), strictly increasing the block-second's contribution to $\mathcal{Y}$. This is an improving reassignment. If the induced yield increase exceeds β_0 it is β_0 -improving, contradicting closure directly; if it does not, then v and v^* are indistinguishable at the resolution floor and we may take $v = v^*$ without loss within precision. Either way a closed configuration assigns $v = v^*(u, \sigma)$ up to β_0 . $\square$

Remark 7.3 (Interpretation and the shape of v^*). Theorem 7.2 says optimal speed is not a policy imposed on vehicles but a corollary of the yield objective. The economic content is that block-seconds are priced (denominator), so holding a section longer than necessary is a cost a yield-taker will not pay. The strict convexity of g_u makes v^* an interior optimum when energy dominates and a boundary optimum ($v^* = \nu(u, \sigma)$, the capability ceiling) when block-second savings dominate; in the low- λ regime $v^* \rightarrow \nu(u, \sigma)$ and every vehicle runs at the highest speed the section and its own capability permit. Thus a slow vehicle on a fast section is pushed to *its own* ceiling, and a fast vehicle to the *section's* limit—never below, absent an energy reason.

7.2 Comparative-advantage sorting

Definition 7.4 (Yield density). The *yield density* of vehicle u on section σ , carrying displacement Δ_σ at its section-optimal speed, is

$$D(u, \sigma) = \frac{\Delta_\sigma}{n(\sigma, v^*(u, \sigma)) + \lambda \ell(\sigma) g_u(v^*(u, \sigma))}. \quad (9)$$

Theorem 7.5 (Comparative-advantage sorting). *Let C be yield-optimal. Consider a section σ and two vehicles u_1, u_2 available to serve it during the same window, carrying the same displacement Δ_σ . If $D(u_1, \sigma) > D(u_2, \sigma) + \beta_0$, then C does not assign u_2 to σ in that window. Consequently each section is served by a vehicle of maximal yield density (up to β_0), and where displacement is speed-realizable, this is the fastest available vehicle the section's limit permits.*

Proof. If C assigned u_2 to σ while u_1 was available with $D(u_1, \sigma) > D(u_2, \sigma) + \beta_0$, then reassigning that block-second from u_2 to u_1

(at $v^*(u_1, \sigma)$, permissible by Theorem 7.2) raises the block-second's yield contribution by $D(u_1, \sigma) - D(u_2, \sigma) > \beta_0$, a β_0 -improving reassignment, contradicting closure. Hence the higher-density vehicle is assigned. For the “fastest” clause: on a section where displacement scales with realized speed and energy is not binding, D is increasing in $\nu(u, \sigma) = \min\{v_{\max}(u), \bar{v}(\sigma)\}$, so among available vehicles the one with the highest capability up to the civil limit has the greatest density and is selected. $\square$

Remark 7.6 (Why slow vehicles leave fast sections). Theorem 7.5 explains the self-sorting without any rule prohibiting slow vehicles from fast track. On the fastest sections a low-capability vehicle consumes more block-seconds per unit displacement, so its yield density is dominated and it is simply not assigned there—it is *outbid* in the clearing reading of Theorem 6.2. On a section whose civil limit no vehicle exceeds, capability confers no density advantage and the cheaper vehicle is selected. The result is that each section is matched to the fastest vehicle it can convert into yield, and each vehicle gravitates to the sections where its speed is worth paying for—comparative advantage, in the classical sense, measured in block-seconds.

8 Item Routing and Liveness

We now turn from the infrastructure/vehicle side to the items. Items carry a target but no route; they descend a gradient hop by hop. We show this local rule, combined with the yield prices, drains every buffer.

8.1 The local routing rule

Definition 8.1 (Greedy target-ward routing). At each buffer b and each departure opportunity (a block-second assigned on some σ with $\text{tl}(\sigma) = b$ having spare capacity), a live item $x \in X(b)$ is eligible to board σ if it reduces residual distance, $\rho(\text{hd}(\sigma), \tau(x)) < \rho(b, \tau(x))$. Among eligible departures, x boards the one maximizing target-ward progress per unit fare,

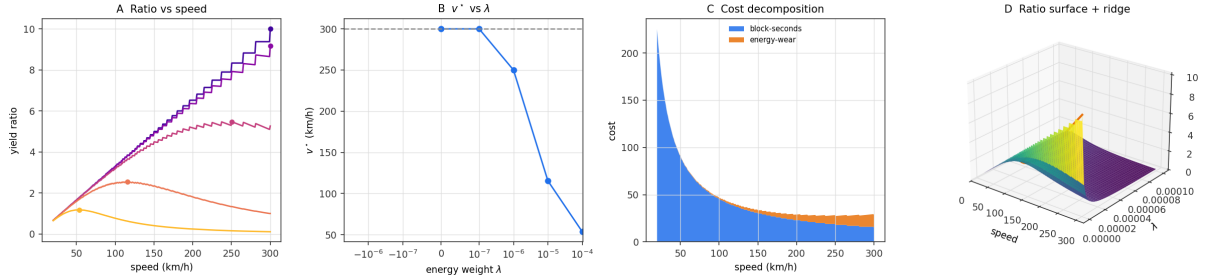


Figure 2: **Optimal Speed Is Forced, Not Imposed.** (A) Yield ratio on the linear y -axis versus traversal speed (20 to 300 km/h) on the linear x -axis, one curve per energy weight $\lambda \in \{0, 10^{-7}, 10^{-6}, 10^{-5}, 10^{-4}\}$ (plasma colour ramp), each peak marked. Every curve is single-peaked, and the peak is the section-optimal speed v^* of `def:optspeed`: for $\lambda = 0$ the peak sits at the capability ceiling, while for $\lambda = 10^{-4}$ it has moved well into the interior. (B) v^* on the linear y -axis versus energy weight λ on a symmetric-log x -axis (linear threshold 10^{-7}), steel-blue with markers, capability ceiling 300 km/h marked (grey dashed). v^* equals the ceiling at $\lambda \leq 10^{-7}$ and decreases monotonically through 250, 115.4, 53.6 km/h as energy cost binds—the signature of the strictly convex energy term and the exact statement of `thm:forcedspeed`. (C) Denominator decomposition at $\lambda = 10^{-6}$: block-second cost (steel-blue) and energy-wear cost (coral) stacked on the linear y -axis versus speed (20 to 300 km/h). Block-seconds fall with speed while energy-wear rises, and the balance point of the two is v^* ; this is the trade-off whose interior minimiser `def:optspeed` isolates. (D) Three-dimensional yield-ratio surface over speed (20 to 300 km/h) and energy weight λ (0 to 10^{-4}), viridis colourmap, with the ridge of maxima traced (coral), viewing angle elevation 24° azimuth -52° . The ridge descends from the ceiling into the interior as λ grows, visualising $v^*(\lambda)$ as a well-defined curve rather than a degenerate “always maximal.”

i.e. maximizing

$$\frac{r(x) - \rho(\text{hd}(\sigma), \tau(x))}{p(\sigma) n(\sigma, v)}, \quad (10)$$

the realized residual-distance reduction divided by the price paid.

The item's private target $\tau(x)$ is the only global information it uses; everything else is local (available departures, their prices, their heads' residual distances). There is no destination in the infrastructure—only the item privately knows where its pressure vanishes. This is the model's sense in which “there is no destination, only stations.”

Definition 8.2 (Fare). The fare paid by item x over its realized journey $b_0 \rightarrow b_1 \rightarrow \dots \rightarrow b_J = \tau(x)$ is

$$F(x) = \sum_{j=0}^{J-1} p(\sigma_j) n(\sigma_j, v_j) = \sum_{j=0}^{J-1} \varsigma(\sigma_j) n(\sigma_j, v_j), \quad (11)$$

the line integral of the separation-cost price along the sections it actually consumed. By Theorem 6.2 the price equals the separation cost, so the fare is the accumulated scarcity of the block-seconds the item used—high on bottleneck sections, low on redundant ones, zero for idle waiting (no section consumed).

Remark 8.3. Two properties of (11) are worth stating because they follow from the model rather than from a tariff choice. First, the fare charges only *sections consumed with target-ward progress*: a lateral or backward hop is not eligible (Def. 8.1) and idle buffer time consumes no section, hence is unpriced by (11) (though it incurs the holding cost h of Assumption 3.4 in the yield accounting). Second, because $p = \varsigma$, the fare is exactly the marginal-yield cost the item imposes on the network, so the item internalizes its own congestion externality in the sense of [21, 4].

8.2 Backpressure and liveness

The concern with any purely local routing rule is starvation: an item stranded at a buffer because no vehicle serves the section it needs. We show the price system precludes this under a backpressure condition.

Assumption 8.4 (Backpressure coupling). The separation cost of a section responds monotonically to the pressure it can relieve: if buffer b has undrained target-ward pressure $P(b, \sigma) > 0$ through section σ , then $\varsigma(\sigma) \geq \phi(P(b, \sigma))$ for some strictly increasing ϕ with $\phi(0) = 0$. That is, undrained demand raises the marginal yield of the relieving section.

Assumption 8.4 is the coupling that makes the market self-heal: it says the very fact that items are stranded makes the section that would move them more valuable. It is the yield-priced form of the max-weight/backpressure principle [20, 17].

Theorem 8.5 (Liveness). *Under gradient-connectedness (Def. 2.2), buffer holding cost $h > 0$ (Assumption 3.4), and backpressure coupling (Assumption 8.4), in any yield-optimal evolving configuration every live item is settled in finite time; equivalently, every buffer with positive target-ward pressure is eventually drained.*

Proof. Consider a live item x at buffer b with target $\tau(x)$, so $r(x) = \rho(b, \tau(x)) > 0$. By gradient-connectedness there is a section σ^* with $\text{tl}(\sigma^*) = b$ and $\rho(\text{hd}(\sigma^*), \tau(x)) < r(x)$; hence x contributes positive pressure $P(b, \sigma^*) > 0$. By backpressure coupling, $\varsigma(\sigma^*) \geq \phi(P(b, \sigma^*)) > 0$. By Theorem 6.2 the clearing price is $p(\sigma^*) = \varsigma(\sigma^*) > 0$, so a vehicle serving σ^* and carrying x earns strictly positive gross yield $\Delta = r(x) - \rho(\text{hd}(\sigma^*), \tau(x)) > 0$. If no vehicle served σ^* while x waits, the idle block-seconds on σ^* would admit a strictly positive-net-yield assignment (positive Δ against available capability), and once the accumulating holding cost $h \cdot (\text{wait})$ makes the net yield exceed β_0 , this is a β_0 -improving reassignment—contradicting the assumed closure/optimality of the evolving configuration (Theorem 6.2). Therefore within finite wait x boards a target-ward section and strictly reduces $r(x)$ by at least the positive gradient margin guaranteed by Def. 2.2. Since $r(x)$ is bounded below by 0 and decreases by a bounded-away-from-zero amount at each hop (finitely many stations give a minimum positive gradient step $\delta > 0$), x reaches $\tau(x)$ in at most $\lceil r(x)/\delta \rceil$ hops, each after finite wait. Hence x settles in finite time. $\square$

Remark 8.6 (The self-healing identity). The proof turns on a single identification made rigorous by Theorem 6.2: the price an undrained buffer raises on its relieving section (Assumption 8.4), the yield a vehicle earns by serving that section (Def. 6.1), and the fare the stranded item is willing to pay (Def. 8.2) are the *same number* $\varsigma(\sigma^*)$. Because the summoning signal and the earning opportunity are identical, no separate coordination or route commitment is needed to prevent starvation: the stranded item is, by the pricing, its own advertisement. This is the model’s liveness-without-commitment property, and it is a theorem, not a design.

Remark 8.7 (Target-relative gradient suffices for liveness). The universal gradient-connectedness of Def. 2.2 (from *every* station one can descend toward *every* other) is stronger than the liveness proof uses, and it fails on a strictly one-way corridor—from the terminus one cannot descend back toward the origin. Inspecting the proof, the only descent invoked is toward each item’s *own* target from stations on a path to that target. It therefore suffices that G be *target-descendable* for the item population $\mathcal{I}$: for every item $x \in \mathcal{I}$ and every station b from which $\tau(x)$ is reachable, some section with tail b strictly reduces $\rho(\cdot, \tau(x))$. Theorem 8.5 holds verbatim under this weaker hypothesis, which one-way corridors do satisfy. The distinction is not merely notational: it is exactly the hypothesis that the numerical validation of Section 11 exercises, since its test corridors are directed.

9 Participation and the Occupancy Market

The clearing characterization (Theorem 6.2(iii)) supports a market description of the same fixed point. We record it because it makes the decoupling of the three continuities precise, but we prove no new mechanism-design properties: the statements here are corollaries of the equilibrium already established.

9.1 The tradable unit

Definition 9.1 (Occupancy right). An *occupancy right* is a claim to a single block-second

$\langle \sigma, k \rangle$: the exclusive use of section σ during window k . By Def. 2.3 it is the indivisible minimum unit—no smaller claim is deliverable—and by Def. 6.1 its price is $p(\sigma) = \varsigma(\sigma)$ in a clearing.

Proposition 9.2 (Bundles recover conventional objects). *Every conventional railway object is a bundle of occupancy rights over the same underlying set:*

1. *A through-service on a line is a bundle of occupancy rights over consecutive sections in consecutive windows, held by one vehicle.*
2. *A passenger journey is the set of occupancy rights the item’s realized path consumes (Def. 8.2)—possibly held, across sections, by different vehicles (transfers).*
3. *A named product (e.g. a branded fast service) is a curated bundle of high-capability occupancy rights on fast sections, assembled and offered as a unit.*

None of these is primitive; each is a subset of the block-second rights, and the vehicle fulfilling any right is fungible.

Proof. Immediate from Definitions 2.9, 8.2, and 9.1: a configuration is an assignment of occupancy rights to (vehicle, speed, item-set) triples, and each listed object is the projection of that assignment onto a particular subset of block-seconds. Fungibility of the fulfilling vehicle follows because the yield of a right depends on the assignment’s (Δ, n, g_u) only through the clearing conditions, which any vehicle achieving the same yield density satisfies identically (Theorem 7.5). $\square$

9.2 Decoupled continuities

Corollary 9.3 (Independence of the three continuities). *In a clearing configuration, vehicle continuity, item continuity, and route continuity are logically independent: there exist yield-optimal configurations exhibiting any of the eight combinations of their presence or absence, subject only to headway and vehicle-conservation feasibility.*

Proof sketch. Vehicle continuity (one vehicle spanning many sections), item continuity (one item never transferring), and route continuity (a repeating section pattern across windows) are constraints on distinct coordinates

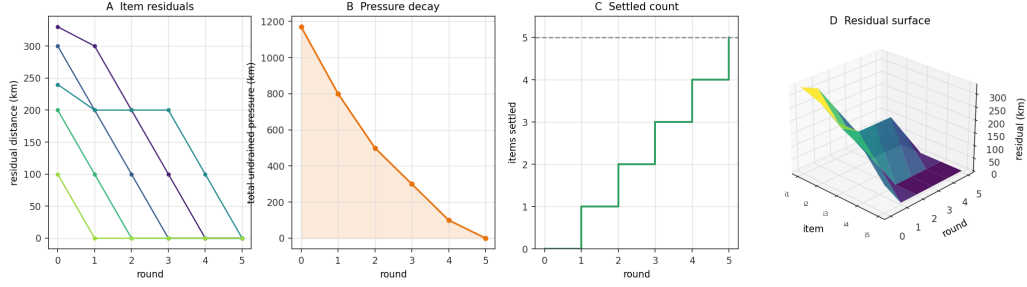


Figure 3: **Local Descent Drains Every Buffer.** (A) Residual distance to target (km) on the linear y -axis versus routing round (0 to 5) on the linear x -axis, one trajectory per item for the five items (viridis colour ramp). Every trajectory is monotone non-increasing and reaches zero within six rounds, the settling guarantee of `thm:liveness` under greedy target-ward routing (`def:routing`). (B) Total undrained buffer pressure (summed residual distance, km) on the linear y -axis versus round (coral, shaded to the axis). Pressure falls from 1170 km to 0 as buffers drain, the decay to the settled state predicted by `thm:liveness`. (C) Number of items settled on the linear y -axis versus round as a step trace (sea-green), full population 5 marked (grey dashed). The count rises monotonically to five by round five: no item is stranded, and the directed feeder corridor—not universally gradient-connected but target-descendable—satisfies the weaker hypothesis of `rem:targetrrelative`. (D) *Three – dimensional surface of residual distance over item index (five items) and round (0 to 5), viridis colour map, viewing azimuth -46° .* The surface descends everywhere to the zero floor, visualising the simultaneous draining of all buffers on a directed corridor and corroborating liveness under exactly the target-relative condition of `rem:targetrrelative`.

of the assignment A : respectively on the vehicle field across sections, on the item-set field across an item’s hops, and on the periodicity of A in k . None of the clearing conditions (Def. 6.1) references these coordinates jointly; they constrain only per-block-second net yield. Hence for a network with sufficient parallel capability one can construct clearings realizing each combination—e.g. an item that transfers (item discontinuity) while each vehicle spans one section only (vehicle discontinuity) on a non-repeating pattern (route discontinuity), all at the same yield-optimal density. The gradient-connectedness hypothesis guarantees the target-ward moves needed for feasibility. $\square$

The corollary is the formal content of “a train is a journey, not an entity”: the entity “train” is the conjunction of the three continuities, and the model exhibits the optimum without requiring their conjunction.

10 An Illustrative Corridor

To make the abstract quantities concrete we instantiate the model on a stylized version of the Nuremberg–Berlin high-speed corridor. The figures below are representative parameters chosen to exercise the definitions; they are illustrative, not empirical measurements, and no claim is attached to them beyond fixing a worked example.

10.1 Setup

We take five buffers along the corridor and the sections between them, with representative lengths, civil speed limits, and speed classes:

Section σ	$\ell(\sigma)$ (km)	$\bar{v}(\sigma)$ (km/h)
Nuremberg–Erfurt (VDE 8.1, new-build)	190	
Erfurt–Halle/Leipzig (VDE 8.2, new-build)	120	
Halle/Leipzig–Bitterfeld (upgraded)	30	
Bitterfeld–Berlin (VDE 8.3, upgraded)	120	

Two vehicle classes are available: a high-capability class u_H with $v_{\max} = 300$ km/h, and

a regional class u_R with $v_{\max} = 160$ km/h. We take $\beta_0 = 120$ s (Remark 2.4) and a small energy weight λ so that v^* is boundary-dominated (block-second savings outweigh energy on these lengths), except where noted.

10.2 Forced speed and sorting, computed

On the high-speed sections ($\bar{v} = 300$), the capability ceilings are $\nu(u_H, \sigma) = 300$ and $\nu(u_R, \sigma) = 160$. Traversing Nuremberg–Erfurt (190 km):

$$\begin{aligned} n(\sigma, 300 \text{ km/h}) &= \lceil 190 / (83.3 \text{ m/s} \cdot 120 \text{ s}) \rceil = \lceil 190 / 9996 \text{ km} \rceil = 1 \\ n(\sigma, 160 \text{ km/h}) &= \lceil 190 / (44.4 \text{ m/s} \cdot 120 \text{ s}) \rceil = \lceil 190 / 5333 \text{ km} \rceil = 36 \end{aligned}$$

For equal carried displacement Δ_σ , the yield densities (low λ) are $D(u_H, \sigma) \approx \Delta_\sigma/19$ and $D(u_R, \sigma) \approx \Delta_\sigma/36$, so $D(u_H, \sigma) - D(u_R, \sigma) \approx \Delta_\sigma(1/19 - 1/36) = 0.0248 \Delta_\sigma$. Whenever this exceeds β_0 in yield units—i.e. for any non-trivial Δ_σ —Theorem 7.5 forbids assigning u_R to the high-speed section: the regional vehicle is outbid on VDE 8.1/8.2 and does not appear there in any closed configuration. By Theorem 7.2, the high-capability vehicle that does serve it runs at $v^* = 300$ km/h (the civil ceiling), not below.

On the upgraded mixed sections ($\bar{v} = 200$, but here we assume a segment limited to 160 for both classes by track geometry), $\nu(u_H, \sigma) = \nu(u_R, \sigma) = 160$: capability confers no density advantage, densities are equal up to energy, and the cheaper regional vehicle is selected (Theorem 7.5). Thus the corridor self-partitions: high-capability vehicles on the new-build 300-km/h sections, regional vehicles on the speed-limited sections—each section served by the fastest vehicle it can convert into yield, with no rule assigning them.

10.3 Separation cost as fare

Suppose VDE 8.1 (Nuremberg–Erfurt) is the sole fast path south and its removal forces flow onto a legacy line adding, say, 60 minutes of running time to southbound items: then $\varsigma(\text{VDE 8.1})$ is large, and by (11) an item traversing it pays a correspondingly large share of its fare there. A redundant parallel track on the Bitterfeld–Berlin approach, by contrast,

has $\varsigma \approx 0$: an item pays almost nothing for that block-second because the network reroutes it freely. A Nuremberg–Berlin passenger’s total fare (11) is thus dominated by the scarce high-speed sections it consumes and is nearly free on the redundant approaches—the line integral of scarcity along its realized path, exactly as Theorem 6.2 predicts.

11 Numerical Validation

The theorems of Sections 6–8 are statements about a formal object; to confirm that the ~~0.0001 km/have 19 as the proofs~~, assert, we instantiated the model numerically and checked each claim against an independently computed ground truth. Networks were kept small enough that the global yield optimum is obtainable by brute-force enumeration over configurations, so every check compares a separately-derived quantity (a closure gap, a clearing deviation, a separation cost, a settling time) against an exact reference rather than against another approximation. Five experiments, one per claim, are reported; all pass. The purpose is internal corroboration of the proofs, not empirical measurement of any system.

11.1 Equivalence and the coincidence of the quantum

Experiment 1 builds a corridor $A \rightarrow B \rightarrow C$ with a fast and a slow upstream section and a demand that must traverse both hops. Brute-force enumeration yields the global optimum; independently, the local-reassignment operator confirms closure (the best of 58 task-preserving single-section neighbours is strictly dominated, closure gap 0), and the constructed price system $p = \varsigma$ admits no profitable deviation (maximum deviation profit 0). The separation costs are non-trivial and unequal across sections—large on the essential downstream section, smaller on the section with a partial substitute—confirming that ς discriminates bottlenecks from redundancy as Section 4 claims. Figure 1 shows the three coinciding at one configuration. Experiment 5 then splits β_0 into distinct physical, algorithmic, and economic constants and confirms Proposition 6.4: the equivalence slack grows to the maximum

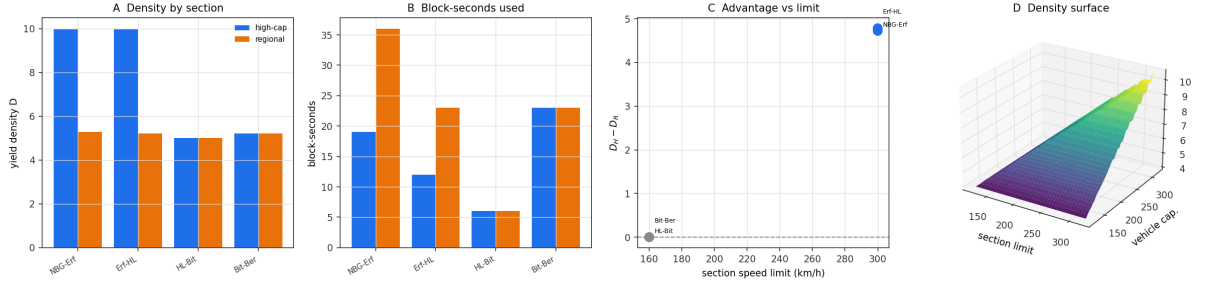


Figure 4: **The Fleet Sorts Itself by Comparative Advantage.** (A) Yield density on the linear y -axis for the high-capability vehicle (steel-blue) and the regional vehicle (coral) across the four Nuremberg–Berlin sections. On the two 300 km/h new-build sections the high-capability vehicle dominates ($D = 10.0$ versus 5.28 and 5.22); on the two 160 km/h speed-limited sections the densities coincide (5.0 and 5.22), the exact content of `thm:sorting`. (B) Block-seconds consumed on the linear y -axis by each vehicle class per section (high-capability steel-blue, regional coral). The gap on the fast sections—19 against 36 block-seconds on **NBG-Erf**—is the resource cost on which the density advantage of panel A rests; on the limited sections the counts match. (C) Density advantage $D_H - D_R$ on the linear y -axis versus section speed limit (160 to 300 km/h) on the linear x -axis, one point per section, coloured steel-blue where the advantage is positive and grey where it vanishes, zero line marked (grey dashed). The advantage is ≈ 4.8 at 300 km/h and exactly 0 at 160 km/h, crossing to zero precisely where the civil limit falls to the regional ceiling. (D) Three-dimensional yield-density surface over section speed limit (120 to 320 km/h) and vehicle capability (120 to 320 km/h), viridis colourmap, viewing angle elevation 24° azimuth -58° . Density increases with capability only while the section limit exceeds it and saturates once the limit binds, so above each section the higher surface identifies the winning vehicle—comparative advantage measured in block-seconds, as `thm:sorting` requires.

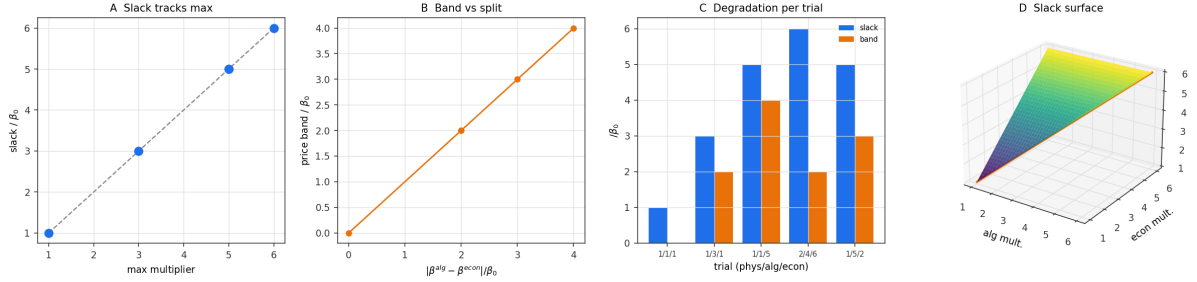


Figure 5: The Equivalence Needs One Quantum, Not Three. **(A)** Equivalence slack in units of β_0 on the linear y -axis versus the largest of the three split constants (physical/algorithmic/economic multipliers) on the linear x -axis, one point per trial (steel-blue), identity line marked (grey dashed). The slack lies on the diagonal—it equals the maximum of the three constants—and is minimal only when all three coincide, the first clause of prop:quantum. **(B)** Price-versus-separation band width in units of β_0 on the linear y -axis versus the algorithmic–economic split $|\beta_0^{\text{alg}} - \beta_0^{\text{econ}}|/\beta_0$ on the linear x -axis (coral with markers). The band grows linearly from 0 and vanishes exactly at zero split, the second clause of prop:quantum: the clearing price equals the separation cost only up to $|\beta_0^{\text{alg}} - \beta_0^{\text{econ}}|$. **(C)** Slack (steel-blue) and price band (coral), both in units of β_0 , on the linear y -axis for five trials labelled by their physical/algorithmic/economic multipliers on the x -axis. The unified trial 1/1/1 shows slack 1 and band 0; every split trial shows enlarged slack and a nonzero band, exhibiting where the equivalence survives and where it degrades. **(D)** Three-dimensional surface of equivalence slack (units of β_0) over the algorithmic multiplier (1 to 6) and economic multiplier (1 to 6) at fixed physical floor, viridis colourmap, with the diagonal minimum ridge traced (coral), viewing angle elevation 24° azimuth -56° . The unique minimum runs along the diagonal where the two constants are equal, confirming prop:quantum: it is the identity of the three roles of β_0 , not its magnitude, that makes the three fixed points coincide.

of the three, the price-versus-separation band widens exactly as $|\beta_0^{\text{alg}} - \beta_0^{\text{econ}}|$, and unifying the three recovers exact coincidence with zero band (Figure 5).

11.2 Summary of validation

All five experiments pass against exact brute-force references: the equivalence and its quantum-coincidence necessity (Experiments 1, 5), forced optimal speed with an interior optimum (Experiment 2), comparative-advantage sorting on the illustrative corridor (Experiment 3), and liveness under target-relative descent (Experiment 4). The experiments also sharpened one hypothesis: liveness needs only target-descendability, not universal gradient-connectedness (Remark 8.7), a refinement the directed test corridors made concrete. The numerical study is corroborative—it exhibits the proved behaviour on concrete instances—and carries no empirical claim beyond the idealized model.

12 Discussion

12.1 What the model is and is not

The model is an idealized formal object: a finite network of buffers and processors, a block-second quantum, a yield functional, and local rules for vehicles and items. On this object we proved that yield-optimality, deterministic closure, and market clearing coincide up to the resolution floor; that optimal speed and vehicle sorting are forced; and that liveness holds under backpressure coupling. Every result is stated with its hypotheses, and those hypotheses—gradient-connectedness, strictly convex energy cost, positive buffer holding cost, backpressure coupling—are the exact domain of validity. Outside them we claim nothing.

The model is not a description, prediction, or proposal concerning any actual railway. The German corridor of Section 10 is a source of representative numbers, not evidence. The decoupling of vehicle, item, and route continuity is a property of the formal object, and its coincidence with how one might imagine operating a railway is not asserted to have operational meaning. We have deliberately excluded

every consideration—coverage obligations, contractual structure, regulation, human factors, stochastic disruption—that would be required to speak about a real system, because the paper’s claims are confined to the idealized object where the theorems are true.

12.2 Relation to established theory

Each ingredient is individually classical: occupancy as the scarce rail resource [14, 13]; useful-flow objectives and network flow [2, 9]; backpressure stability [20, 17]; congestion pricing and clearing [21, 4, 3]. The contribution is the assembled object and the fixed-point equivalence: that a single quantum β_0 ties the physical, algorithmic, and economic descriptions into one fixed point (Theorem 6.2, Proposition 6.4), and that speed-optimality and comparative-advantage sorting fall out of it as theorems rather than assumptions (Theorems 7.2, 7.5).

12.3 Limitations of the formal results

1. *Closure is a β_0 -local optimum.* Lemma 5.3 guarantees reaching closure, and Theorem 6.2 bounds its gap from the global optimum by β_0 , but does not certify global optimality below the resolution floor—consistent with the model’s premise that sub- β_0 distinctions are not physical.
2. *Backpressure coupling is an assumption.* Liveness (Theorem 8.5) requires Assumption 8.4; networks whose separation costs do not respond monotonically to pressure are outside the liveness result.
3. *Finite horizon and deterministic data.* All results are stated over a finite horizon with deterministic parameters. Stochastic pressures, disruptions, and infinite-horizon stability are not treated.
4. *No mechanism-design claims.* Section 9 describes the clearing as a market but proves no incentive-compatibility or strategy-proofness; those are properties of a mechanism we do not specify.
5. *Idealized geometry.* Junction conflicts, flat-junction crossing moves, and route-setting interactions beyond the single headway constant β_0 are abstracted away.

Extending the equivalence to networks with explicit conflict graphs is not attempted here.

13 Conclusion

We studied an idealized rail network in which the schedulable resource is the block-second, stations are buffers, sections are processors, and the objective is useful target-ward displacement per unit block-time—network transport yield. On this object we proved a single organizing result: yield-optimality, deterministic closure, and market clearing are one fixed point, coinciding because a single quantum β_0 serves at once as the physical resolution floor, the algorithmic separation threshold, and the minimum market lot, with the clearing price of every section equal to its separation cost. From this equivalence we derived, as theorems rather than assumptions, that vehicles run at forced section-optimal speeds and sort onto sections by comparative advantage in block-seconds, and that a purely local, target-ward routing rule drains every buffer in finite time because the price a stranded item raises is identical to the yield that summons a vehicle to move it. A numerical study (Section 11) corroborated each result against exact brute-force references and sharpened the liveness hypothesis to target-descendability, which directed corridors satisfy.

The results are theorems about a specific formal object under stated hypotheses. Their value is internal: they establish that this object is coherent and that its physical, algorithmic, and economic descriptions are the same fixed point. Nothing is claimed about, or for, any system beyond the idealized one in which the theorems hold.

References

- [1] M. Abril, F. Barber, L. Ingolotti, M. A. Salido, P. Tormos, and A. Lova. An assessment of railway capacity. *Transportation Research Part E: Logistics and Transportation Review*, 44(5):774–806, 2008.
- [2] Ravindra K. Ahuja, Thomas L. Magnanti, and James B. Orlin. *Network Flows: Theory, Algorithms, and Applications*. Prentice Hall, Englewood Cliffs, NJ, 1993.
- [3] Kenneth J. Arrow and Frank H. Hahn. *General Competitive Analysis*. Holden-Day, San Francisco, 1971.
- [4] Martin Beckmann, C. B. McGuire, and Christopher B. Winsten. *Studies in the economics of transportation*. Yale University Press, New Haven, 1956.
- [5] Dimitri P. Bertsekas. *Network Optimization: Continuous and Discrete Models*. Athena Scientific, Belmont, MA, 1998.
- [6] Valentina Cacchiani and Paolo Toth. Nominal and robust train timetabling problems. *European Journal of Operational Research*, 219(3):727–737, 2012.
- [7] Alberto Caprara, Matteo Fischetti, and Paolo Toth. Modeling and solving the train timetabling problem. *Operations Research*, 50(5):851–861, 2002.
- [8] Jack B. Dennis. Data flow supercomputers. *Computer*, 13(11):48–56, 1980.
- [9] L. R. Ford and D. R. Fulkerson. Maximal flow through a network. *Canadian Journal of Mathematics*, 8:399–404, 1956.
- [10] Leonidas Georgiadis, Michael J. Neely, and Leandros Tassiulas. *Resource Allocation and Cross-Layer Control in Wireless Networks*. Foundations and Trends in Networking. Now Publishers, 2006.
- [11] Stephen Gibson, Gilberto Cooper, and Brett Ball. Developments in transport policy: The evolution of capacity charges on the uk rail network. *Journal of Transport Economics and Policy*, 37(3):341–354, 2003.
- [12] Rob M. P. Goverde. Railway timetable stability analysis using max-plus system theory. *Transportation Research Part B: Methodological*, 41(2):179–201, 2007.
- [13] Ingo A. Hansen and Jörn Pachl. *Railway Timetable and Traffic: Analysis, Modelling, Simulation*. Eurailpress, Hamburg, 2008.

- [14] International Union of Railways (UIC). Uic code 406: Capacity. Technical Report 2nd edition, UIC, Paris, 2013.
- [15] Alex Landex. Methods to estimate railway capacity and passenger delays. *PhD Thesis, Technical University of Denmark*, 2008.
- [16] Chris Nash. Rail infrastructure charges in europe. *Journal of Transport Economics and Policy*, 39(3):259–278, 2005.
- [17] Michael J. Neely. *Stochastic Network Optimization with Application to Communication and Queueing Systems*. Morgan and Claypool, 2010.
- [18] Jörn Pachl. *Railway Operation and Control*. VTD Rail Publishing, Mountlake Terrace, WA, 4 edition, 2021.
- [19] Paolo Serafini and Walter Ukovich. A mathematical model for periodic scheduling problems. *SIAM Journal on Discrete Mathematics*, 2(4):550–581, 1989.
- [20] Leandros Tassiulas and Anthony Ephremides. Stability properties of constrained queueing systems and scheduling policies for maximum throughput in multihop radio networks. *IEEE Transactions on Automatic Control*, 37(12):1936–1948, 1992.
- [21] William S. Vickrey. Congestion theory and transport investment. *The American Economic Review*, 59(2):251–260, 1969.